

Phase 3 - Project Design

PocketSmart AI: Your Smart Budget & Recommendation Assistant

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1. Design Overview

PocketSmart AI uses a modular architecture. The frontend collects user information, FastAPI handles routes and sessions, services manage recommendation and Gemini communication, and the database layer supports persistence.

2. Architecture Components

Component	Responsibility
Frontend UI	HTML/CSS + JavaScript/Jinja2 input and result pages
FastAPI Backend	Routing, sessions and application communication
Gemini AI Layer	Budget/context processing and multimodal recommendations
Recommendation Service	Recommendation generation and processing
Catalog Service	Product/service catalog information
Database	Application persistence

3. Design Principle

Presentation, routing, authentication, AI communication, recommendation logic and persistence are separated so each module has a clear responsibility.

Project Design - Module Architecture

4. Main Modules

Module	Purpose
main.py	FastAPI entry point and startup
auth.py	Authentication support
config.py	Configuration
database.py	Database setup/access
models/schemas.py	Input/output schemas
routers/auth.py	Register/login/logout routes
routers/pages.py	Page routes
routers/planners.py	Planner routes
services/catalog.py	Catalog/platform data
services/gemini_utils.py	Gemini utility layer
services/recommendation_service.py	Recommendation processing

5. Planner Design

Home, Party and Jewelry planners use different input fields but share the same overall recommendation architecture. This makes the application modular and extensible.

Project Design - Data Flow and Routes

6. Recommendation Data Flow

- User opens planner.
- Frontend collects budget and context.
- FastAPI receives request.
- Schema/model layer structures input.
- Recommendation service prepares context.
- Gemini utility communicates with AI.
- Catalog/platform information is used where available.
- Structured result is returned to UI.
- Relevant information can be stored for history.

7. Major Routes

Route	Purpose
/generate-home	Home recommendations
/generate-party	Party recommendations
/generate-jewelry	Jewelry recommendations
/recommendations-details	Detailed recommendation information
/history	Past recommendation queries/results
/startup	Application initialization
__main__	Application server entry point

Project Design - UI and Storage

8. Main Pages

- index.html - main interface.
- login.html - authentication.
- register.html - registration.
- dashboard.html - personalized dashboard.
- history.html - recommendation history.
- home_planner.html - home inputs.
- party_planner.html - party inputs.
- jewelry_planner.html - jewelry inputs.
- testimonials.html - user reviews/success stories.

9. Static Layer

static/css/style.css controls presentation and static/js/app.js supports client-side interactions and dynamic result display.

10. Database

The project includes data/pocketsmart.db for application persistence through the database layer.

11. Design Outcome

The design connects three planners through one reusable recommendation architecture while keeping authentication, AI logic, UI and data handling modular.